DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTETC402 - SIGNALS AND SYSTEMS

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 Which of the following is a periodic signal?
A)sin(t) B)e^t C) t^2 D) ln(t)

2 Which operation is used to check linearity of a system?
A) scaling B) additivity C) both a & c D) None

3 If h1, h2 and h3 are cascaded, find the overall impulse response
A)h1 * h2 * h3 B) h1 + h2 +h3 C) h1 - h2 - h3 D) h1 - h2 +h3

4 What is the Laplace Transform of δ(t)?
A) 0 B) 1 C) s D) -1

5 Convolution in time domain is equivalent to _____ in frequency domain.
A) Division B) Addition C) Convolution D) Multiplication

6 The fundamental frequency of a periodic signal with period T is:
A) 2π/T B) 1/T C)T D) T/2

7 The Nyquist rate for a signal with maximum frequency fmax=10 kHz is
A) 5 kHz B)10 kHz C)15 kHz D).20 kHz

8 The DTFT of x[n] = δ[n] is:
A)1 B) 0 C)$e^{-j\omega n}$ D) None

9 The Fourier transform of a constant is:
A) delta function B)sinusoid C) constant D) Zero

10 The Laplace Transform is a generalization of:
A) Fourier Transform B) Z- Transform C) Convolution D) None

11 The convolution of x(t) with δ(t) gives:
A) 0 B) x(t) C) x(t−1) D)x(0)

12 A signal that does not repeat itself over time is:

A) Aperiodic B) even C) periodic D)Odd

Q2. Solve the following.

A) Show that any real signal x(t) can be expressed as the sum of its even and odd components. Also, write the expressions for even and odd parts. 6

B) State the derive the expression for convolution integral for continuous-time signals. Mention its significance. 6

Q3. Solve the following.

A) Explain the following signals with examples 6
i)Energy and Power
ii) Analog and Digital

B) Explain Step , Ramp ,delta signal and write the mathematical relationships between them. 6

Q4. Solve Any Two of the following.

A) If $X(e^{jw})$ is the Fourier Transform of a real sequence X[n] then, show the following: 6
(i) $X(e^{jw})$ is conjugate symmetric
(ii) phase of $X(e^{jw})$ is antisymmetric
(iii) magnitude of $X(e^{jw})$ is symmetric

B) Define Convolution. Also Perform the following convolution, x(t) = u(t), and v(t) $=e^{-t}$ u(t) 6

C) Define DTFT. Write its mathematical expression. State any six properties of DTFT. 6

Q5. Solve Any Two of the following.

A) Find the Fourier transform given signal and sketch the magnitude and phase spectrum for $e^{at} \cdot u(-t)$ 6

B) Define Laplace Transform. Find the Laplace Transform of $x(t)=t.e^{-2t} \cdot u(t)$ 6

C) Determine the z-transform and ROC for $(0.5)^n\{u(n+4)-u(n-5)\}$ 6

Q6. Solve Any Two of the following.

A) Sketch each of the following signals. 6
i) u(n+4)- u(n-5)
ii) r(t) + r(t+2)

B) State and explain the Dirichlet conditions for the existence of Fourier series. 6

C) Find the x[n]*h[n] using tabular method , x[n] = {1,2,3} for n=0:2 and h[n]= {1,2,1,-1} for n=0:3 6

*** * * End * * ***